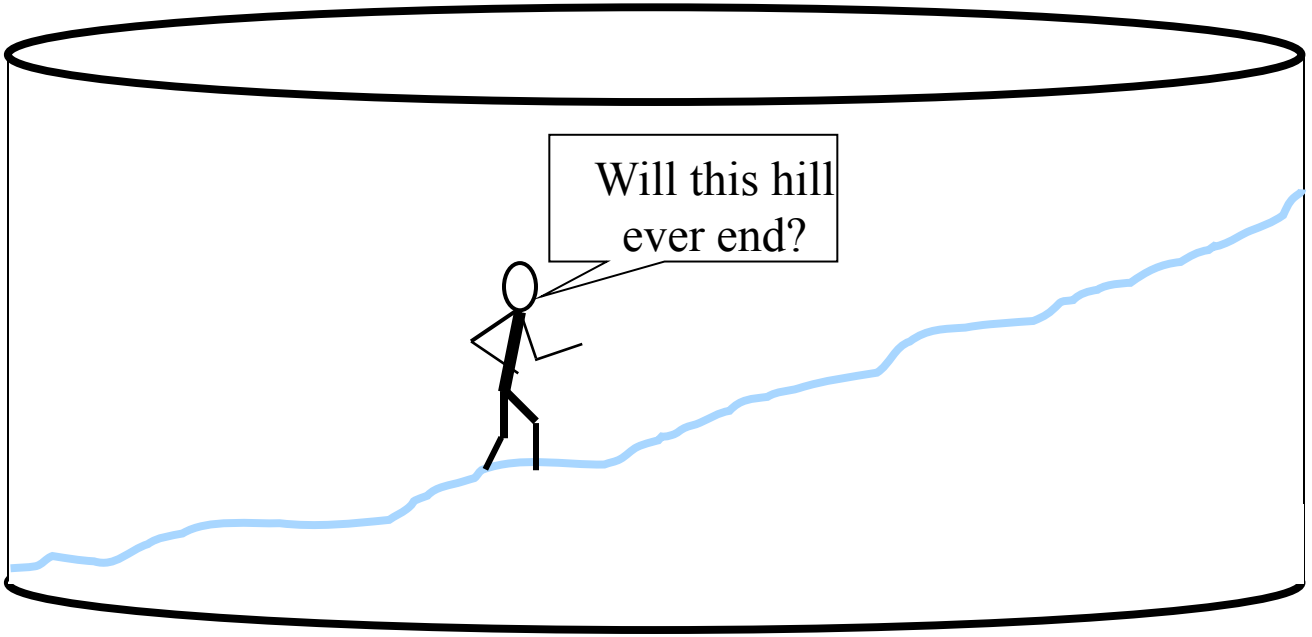
Making Decoherence Run Uphill

(carrying 99 lbs., uphill both ways, in the snow)

Institute for Quantum Information Seminar (1/19/01)



Dave Bacon

Department of Physics and Department of Chemistry

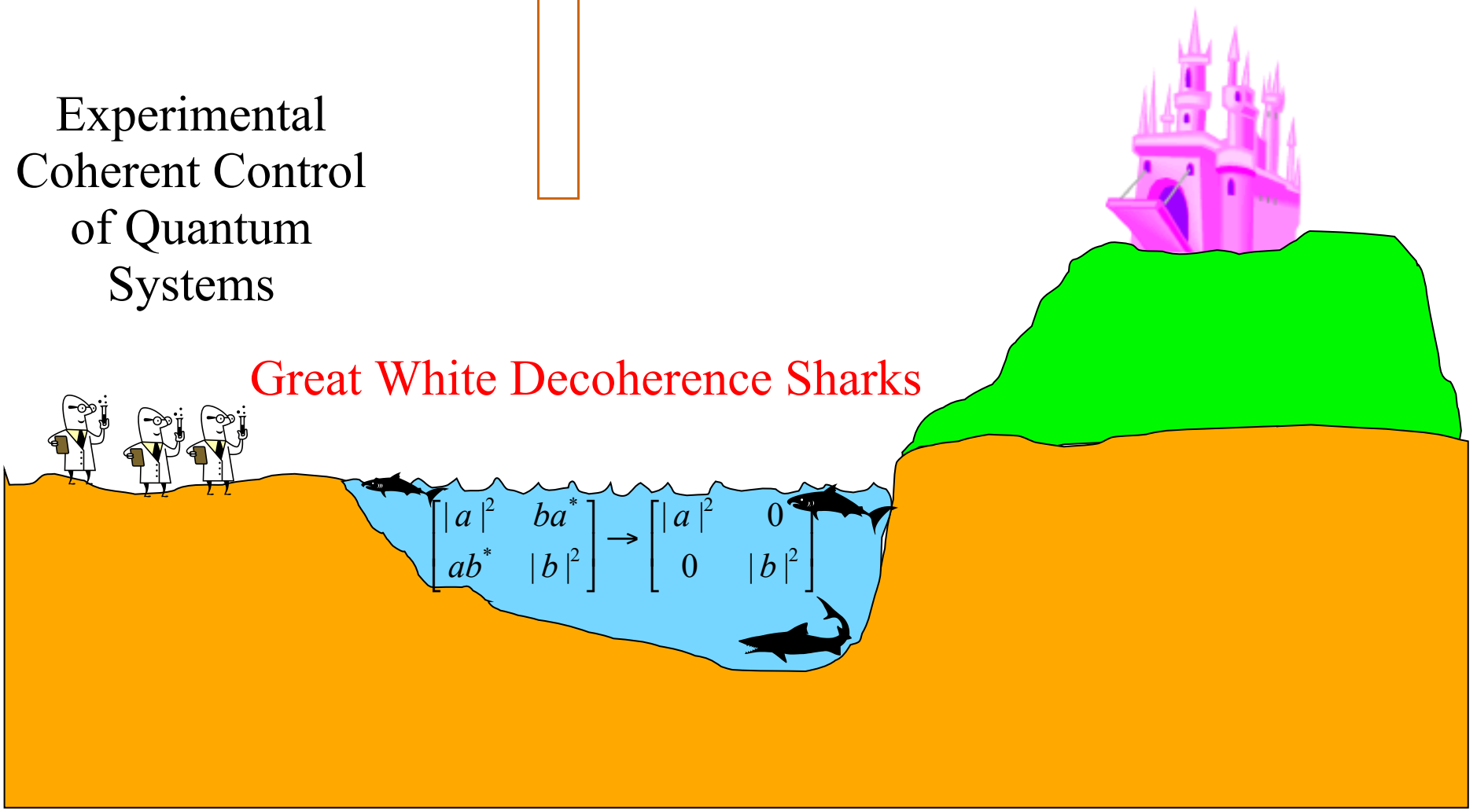
University of California, Berkeley

Fight the Gap

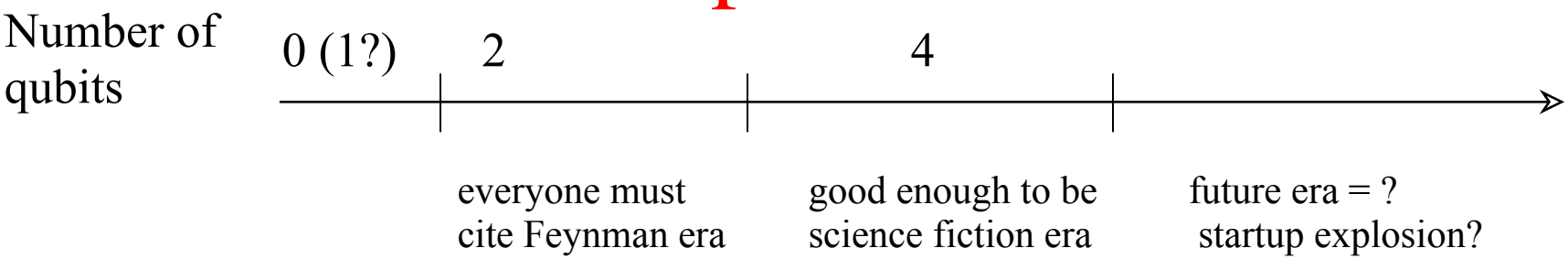
Fault-Tolerant
Quantum Error Correction

Experimental
Coherent Control
of Quantum
Systems

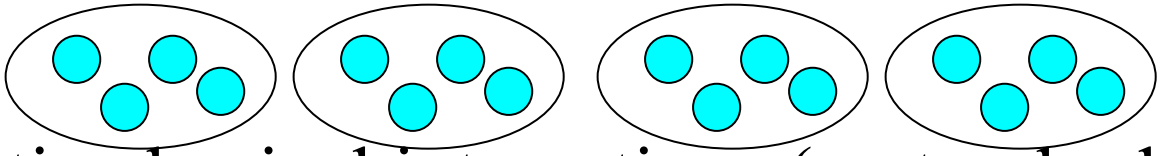
Great White Decoherence Sharks



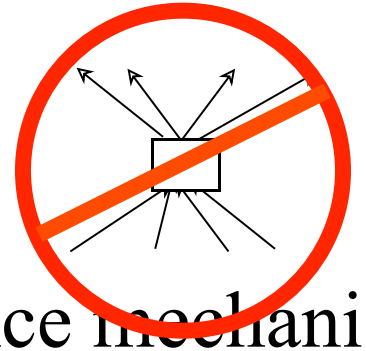
Simpleton

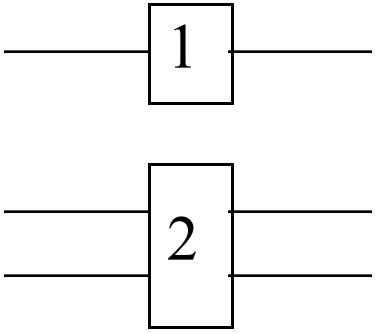


- "few qubit" methods for suppression of decoherence



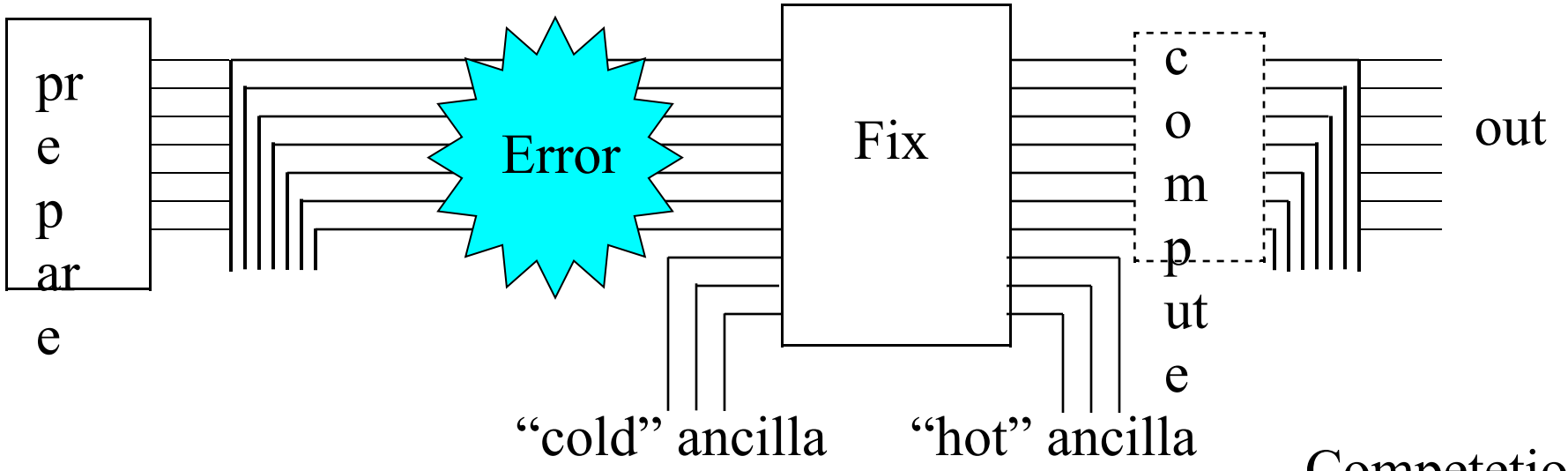
- realistic physical interactions (no ten-body interactions, consider geometry)





- tailor to decoherence mechanisms

Quantum Error Correction Philosophy



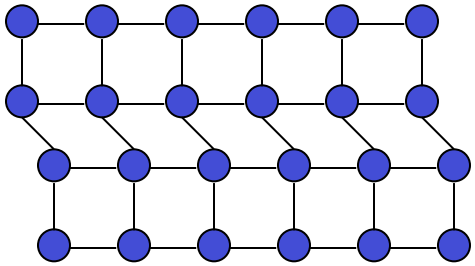
Competetion between errors and cooling

Make It More Natural?

Kitaev's

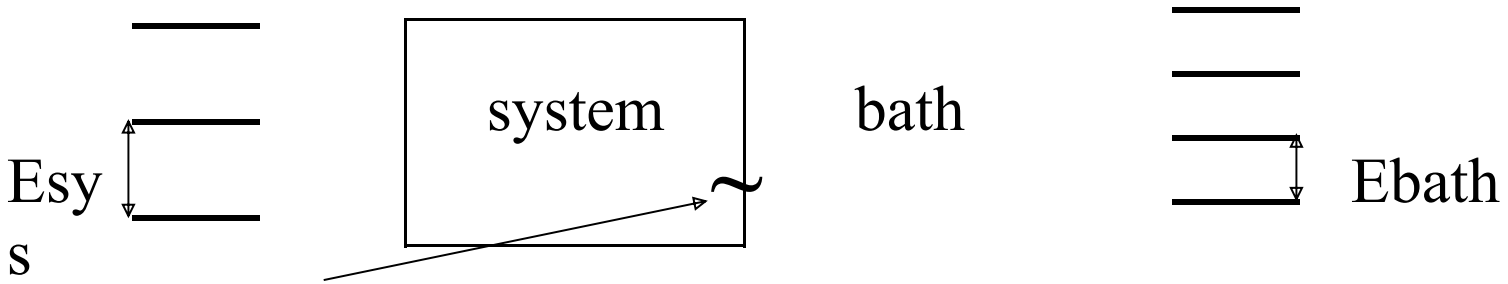
Topological codes.

Majorana Fermions.



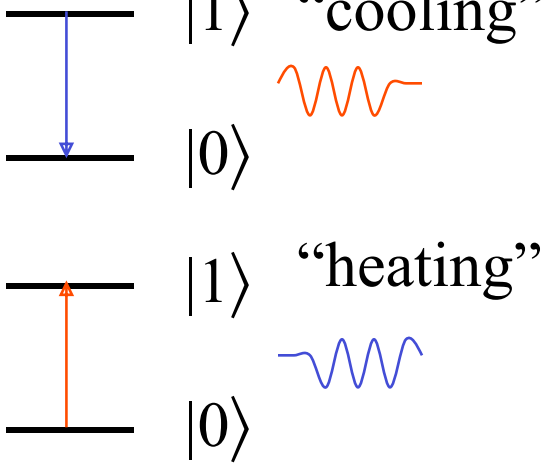
smaller scale realistic versions of these techniques?

Making Decoherence Work Hard



HI = perturbative system-bath coupling

HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies*

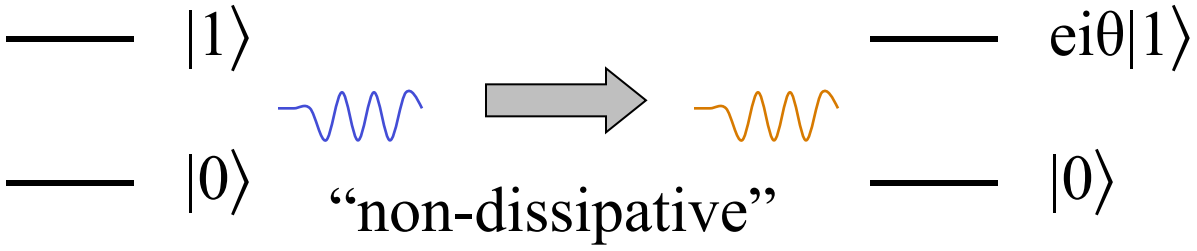


$$\mathbf{H}_I = \eta \sum_k g_k (\sigma_+ + \sigma_-)(\mathbf{a}_k + \mathbf{a}_k^\dagger)$$

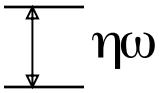
$\not\Leftarrow$

rotating wave approximation

$$\eta \sum_k g_k (\sigma_+ \mathbf{a}_k + \sigma_- \mathbf{a}_k^\dagger)$$



two-level atom radiatively coupled to a thermal reservoir:



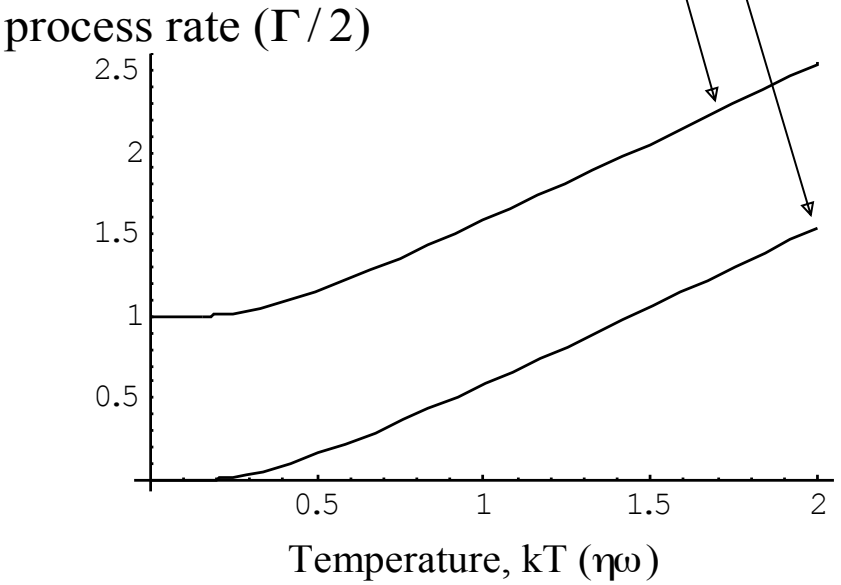
$$\frac{\partial \rho}{\partial t} = -\bar{n}_{th}\frac{\Gamma}{2}\left[\sigma_-\sigma_+\rho + \rho\sigma_-\sigma_+ - 2\sigma_+\rho\sigma_-\right] \quad \text{“heating”}$$

$$-(\bar{n}_{th}+1)\frac{\Gamma}{2}\left[\sigma_+\sigma_-\rho + \rho\sigma_+\sigma_- - 2\sigma_-\rho\sigma_+\right] \quad \text{“cooling”}$$

$$\left\{ \varphi\left[\rho - \sigma_z\rho\sigma_z\right] \right. \quad \text{“non-dissipative”}$$

non-dissipative dephasing

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right) - 1}$$



At low bath temperatures, heating disappears……..

Can We Make All Quantum Errors Run Uphill?

Stabilizer 101(Ph/CS 219)

recall Pauli Stabilizer codes:

"Pauli group *P*" = set of all tensor products of Pauli spin operators ***X***, ***Y***, ***Z*** , ***I*** (±1, ±i) (example element: ***IXXIIIZ***)

"Pauli Stabilizer group *S*" = abelian subgroup of the Pauli group, generated by ***Si***

example: ***S1=XXXX, S2=ZZZZ***

"Code" = simultaneous +1 eigenstates of ***Si*** elements

example:

$$|00\rangle = \frac{1}{\sqrt{2}}\left(|0000\rangle + |1111\rangle\right) \qquad |10\rangle = \frac{1}{\sqrt{2}}\left(|0110\rangle + |1001\rangle\right)$$

$$|01\rangle = \frac{1}{\sqrt{2}}\left(|0011\rangle + |1100\rangle\right) \qquad |11\rangle = \frac{1}{\sqrt{2}}\left(|1001\rangle + |0110\rangle\right)$$

"Normalizer *N*" = Pauli elements which fix *S* under conjugation. (*N*/*S* = logical operators)

normalizer elements = operations on encoded qubits

"Errors which can be detected" anticommute with elements of *S*

example: $S=\{\boldsymbol{I}, \boldsymbol{S1}, \boldsymbol{S2}, \boldsymbol{S1S2}\}$ above: all single qubit errors anitcommute with an element of *S*

$$\langle i \,|\, \mathbf{E} \,|\, j \rangle = \langle i \,|\, \boldsymbol{S}_k \mathbf{E} \,|\, j \rangle = -\langle i \,|\, \boldsymbol{E}\boldsymbol{S}_k \,|\, j \rangle = 0$$

~~Simple~~ Keeping it Real ~~Real~~ Simple

[4,2,2] quantum error detecting code: $\langle jL|Ek|iL\rangle = 0$ (Ek single qubit error)

Stabilizer generators ($Si|iL\rangle = |iL\rangle$)

$S1 = XXXX, \quad S2 = ZZZZ$

Logical operators for 2 encoded qubits ($|iL\rangle = |i1\rangle \otimes |i2\rangle$)

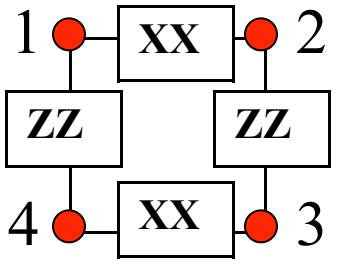
$X1 = XXII, \quad Z1 = IZZI, \quad X2 = IXXI, \quad Z2 = ZZII$

Use to understand the spectrum of the Hamiltonian (stabilizer is a symmetry):

$H0 = XXII + IIXX + IZZI + ZIIZ \qquad ([H0, Si] = 0)$

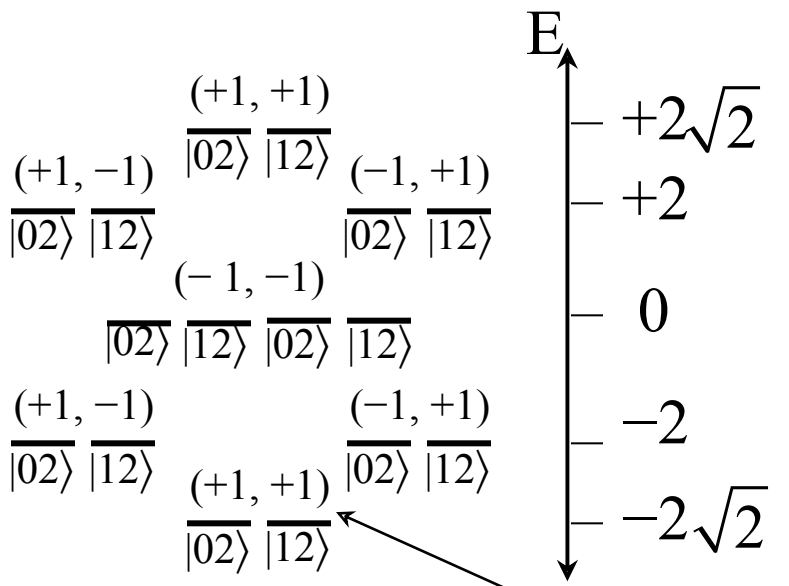
$H0 = X1(I + S1) + Z1(I + S2)$

(2nd encoded qubit not envolved)



S1	*S2*	$H0 = X1(I+S1) + Z1(I+S2)$	eigenvalues (E)
+1	+1	$2(X1+Z1)=2\sqrt{2}H2$	$\pm 2\sqrt{2}$
+1	−1	$2X1$	± 2
−1	+1	$2Z1$	± 2
−1	−1	0	0

$$H \equiv \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



(*S1, S2*) label the stabilizer eigenvalues

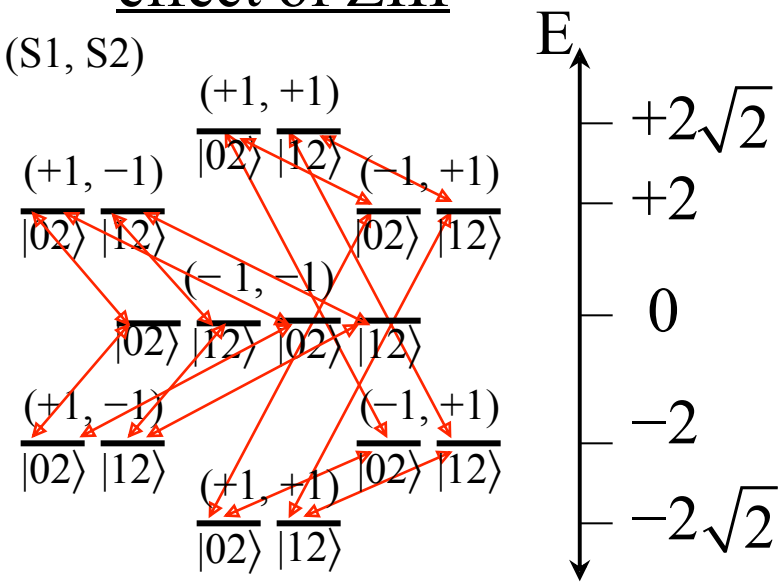
$|02\rangle$, $|12\rangle$ label the second encoded qubit

(note extra degeneracy of (−1, −1))

degenerate ground state:
degeneracy = second encoded qubit

Non-additive code

effect of ZIII



ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

compare to:

two-level nondegenerate system

— $|1\rangle$

— $|0\rangle$

σz: no change of energy

σ-: $|1\rangle$ goes to lower energy

σ+: $|0\rangle$ goes to higher energy

two-level degenerate system

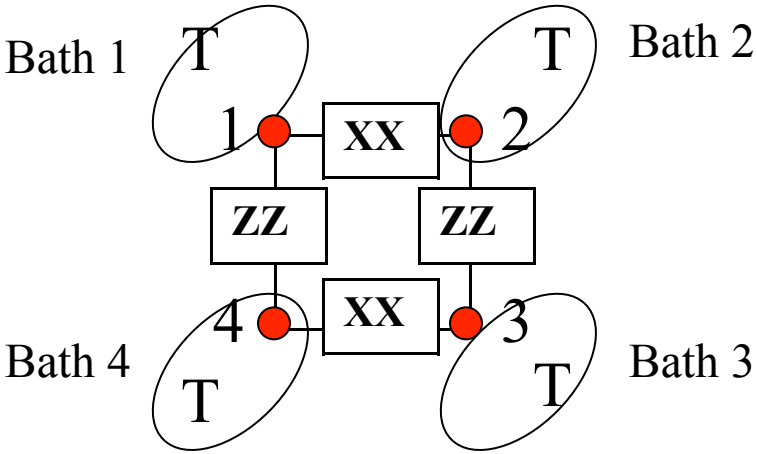
— $|0\rangle$ — $|1\rangle$

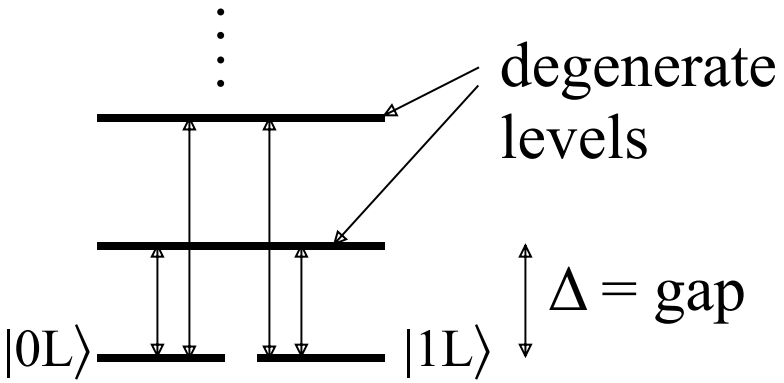
σα: no change of energy

Supercoherence

Ground state = single qubit error detecting code

independent errors…couple to independent baths of temperature T





$$\mathbf{H}_{\text{sys}}(\Delta)=\frac{\sqrt{2}-1}{2}\Delta(\mathbf{XXII}+\mathbf{IIXX}+\mathbf{IZZI}+\mathbf{ZIIZ})$$

$$\mathbf{H}_{\text{bath}} = \sum_{i=\{1,2,3,4\}} \mathbf{H}^{(i)}_{\text{individual bath}}$$

$$\mathbf{H}_I = \sum_{\substack{i=\{1,2,3,4\}\\ \alpha=\{x,y,z\}}} \sigma_\alpha^{(i)} \otimes \mathbf{B}_{\alpha,i}$$

Acts on ith bath

Acts on ith qubit

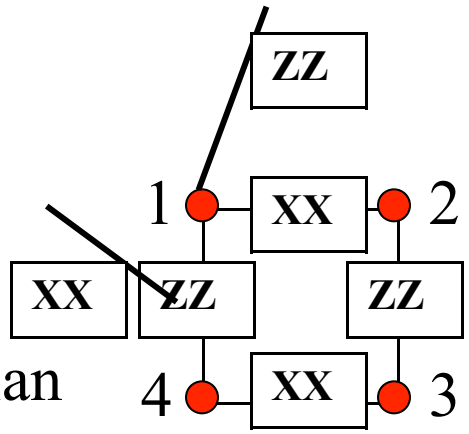
Interaction picture + rotating wave: all single qubit (with individual baths) decoherence processes must be "heating" processes: exponential suppression at low temperatures!

To Compute or Not to Compute

Information encoded in the second logical qubit.

Manipulation of this information can be obtained by control over these two-body interactions:

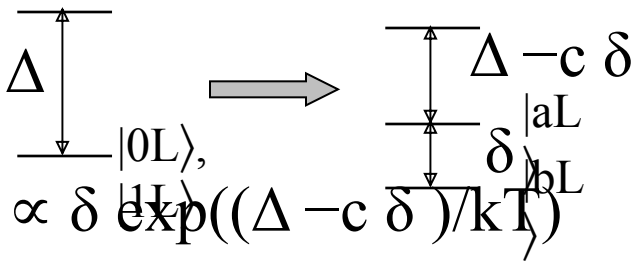
X2=IXXI, ***Z2=ZZII***



control via turning on and off these interactions with strength δ

gap is shrunk by turning on the control Hamiltonian

gap = Δ −c δ, speed of operations = δ,



decoherence rate ∝ exp(−gap/kT)

number of operations in a decoherence time ∝ δ exp((Δ −c δ)/kT)

maximum when δ ∝ kT: number of operations still exponential in T

More Than One Qubit?...

Single qubit angular momentum operators. We will think of a qubit as a spin-1/2 particle.

$$s_x = \tfrac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad s_y = \tfrac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad s_z = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \left[s_\alpha , s_\beta \right] = i\varepsilon_{\alpha\beta\gamma} s_\gamma$$

Total angular momentum operators on n qubits → Collective su(2) operators:

$$\boxed{S_\alpha^n = \sum_{i=1}^{n} s_\alpha^{(i)}} \quad \text{where } s_\alpha^{(i)} = I \otimes \ldots \otimes I \otimes \underbrace{s_\alpha}_{i\text{th qubit}} \otimes I \otimes \ldots \otimes I \qquad \left[S_\alpha^n , S_\beta^n \right] = i\varepsilon_{\alpha\beta\gamma} S_\gamma^n$$

Lie algebra su(2)

$$L = \bigoplus_{J=0,\frac{1}{2}}^{\frac{n}{2}} \bigoplus_{\lambda=1}^{n_J} L_{2J+1}$$

nJ = degeneracy of Jth irrep

2J+1 dimensional irreducible representation of su(2)

J labels irrep

λ labels degeneracy

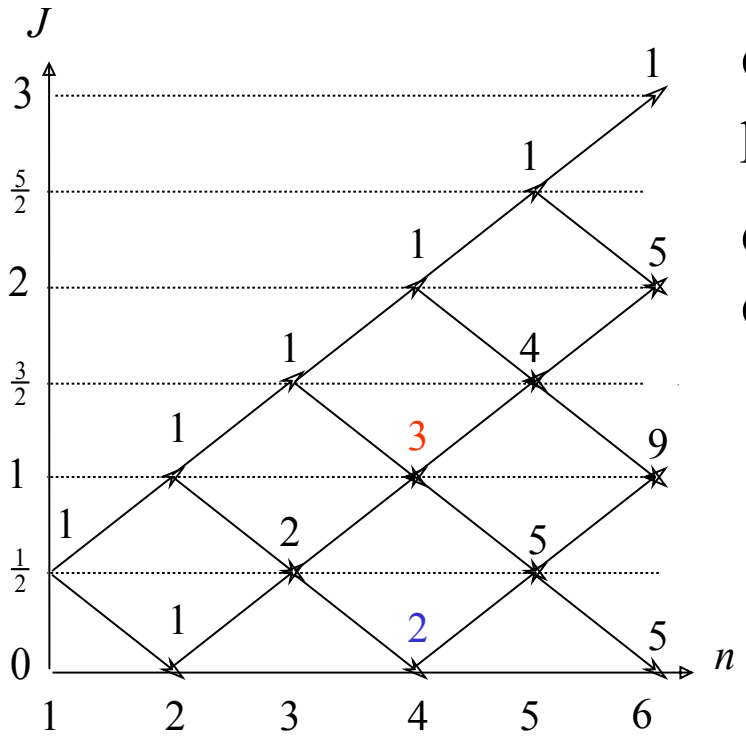
Complete set of commuting observables on n qubits:

$$\underbrace{\left(S^1\right)^2, \left(S^2\right)^2, \left(S^3\right)^2, \ldots, \left(S^{n-1}\right)^2}, \quad \left(S^n\right)^2 \quad , \quad S_z^n$$

labels the degeneracy of an irrep

labels the irrep

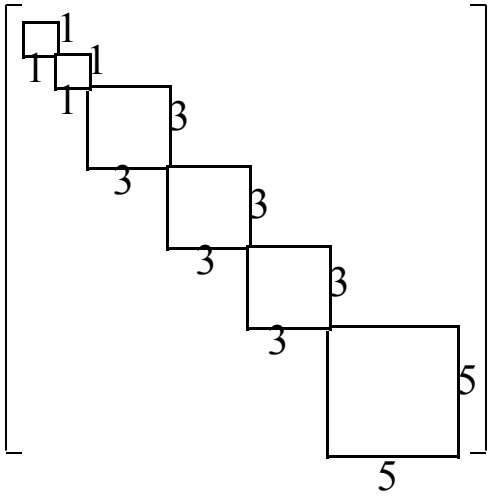
internal quantum number for a particular irrep



create a spin-J particle out of n spin-1/2 particles

degeneracy = different routes to such a construction

$$S_\alpha^4 =$$



Basis: $|J,m,\lambda\rangle$

$$\left(S^n\right)^2 |J,m,\lambda\rangle = J(J+1)|J,m,\lambda\rangle, \quad S_z^n |J,m,\lambda\rangle = m|J,m,\lambda\rangle$$

Information encoded into the degeneracy index λ is immune to collective errors: noiseless (decoherence-free) subsystem

Emanuel Knill, Raymond Laflamme, Lorenza Viola, Phys. Rev. Lett. **84**, 2525 (2000)

Computing on the Degeneracy

Heisenberg coupling between ith and jth qubits:

Heisenberg coupling between qubits can be used to perform any operation on the degeneracy index λ.

Heisenberg = exchange interaction = changes order of addition of angular momentum.

Bacon, Kempe, Lidar, and Whaley, PRL **85** (2000) 1758.
Kempe, Bacon, Lidar, and Whaley, accepted for publication in PRA, quant-ph/0004064

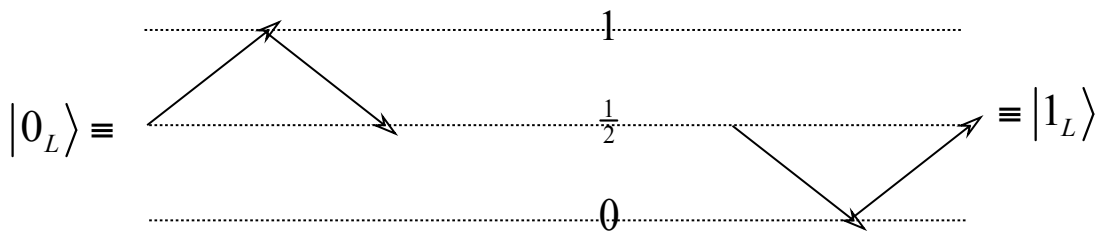
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n=3: the states are formed from combining a triplet with a spin-$\frac{1}{2}$ state or or by combining a singlet with a spin-$\frac{1}{2}$ state.
Triplet is invarient under exchange and singlet flips sign under exchange. Thus

$E_{12} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ on the logical qubits.

more details tell us that

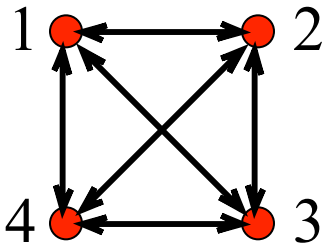
$E_{23} \rightarrow \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ on the logical qubits.



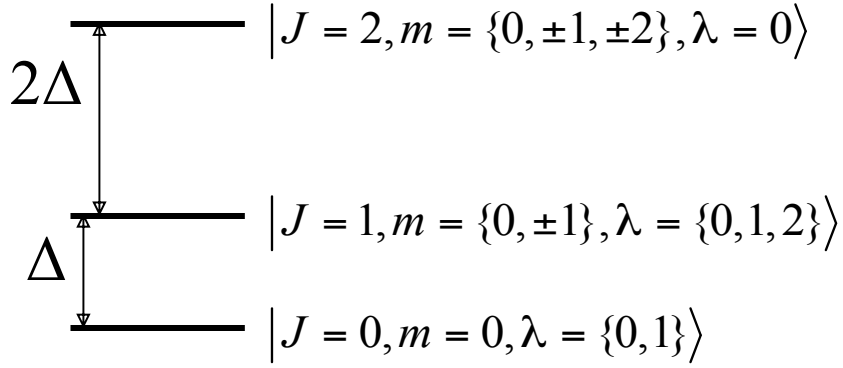
Another Supercoherent Qubit

$$\boldsymbol{H}_0^{(n)}(\Delta) = \tfrac{\Delta}{2}\left(\boldsymbol{S}^{(n)}\right)^2 = \tfrac{\Delta}{2}\left(\sum_{i\neq j=1}^{n} \boldsymbol{H}_{i,j}^{heis} + \tfrac{n}{4}\boldsymbol{I}\right) = \tfrac{\Delta}{2}\left(\sum_{i\neq j=1}^{n} \boldsymbol{s}^{(i)}\boldsymbol{s}^{(j)} + \tfrac{n}{4}\boldsymbol{I}\right)$$

$$\boldsymbol{H}_0^{(n)}(\Delta)\left|J,m,\lambda\right\rangle = \tfrac{\Delta}{2}J(J+1)\left|J,m,\lambda\right\rangle$$



n=4



$$\boldsymbol{S}_\alpha^4\left|J=0,m=0,\lambda=\{0,1\}\right\rangle = 0$$

$$\rightarrow \boldsymbol{GGGG}\left|J=0,m=0,\lambda=\{0,1\}\right\rangle = \left|J=0,m=0,\lambda=\{0,1\}\right\rangle$$

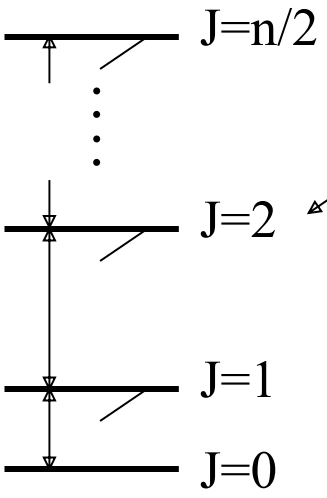
"Supercoherent Quantum Bits"
Bacon, Brown, and Whaley quant-ph/0012018 (2000) submitted to PRL

Ground state J=0 stabilized by collective operations implies

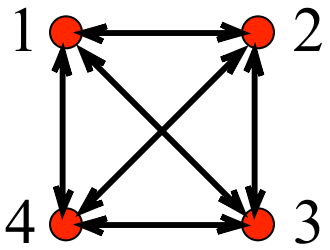
all single qubit operators take states to higher energy levels

Exchange Supercoherent Qubits

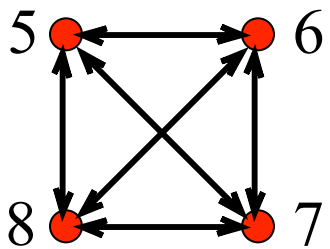
n=even *single qubit allowed transitions*



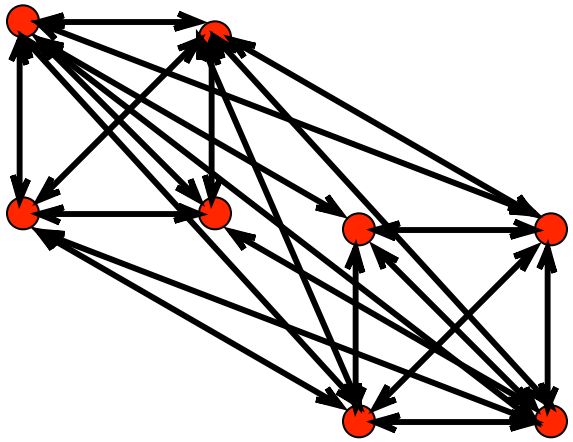
Exchange is universal on the degeneracy of each J irrep



Encoded qubit 1



Encoded qubit 2



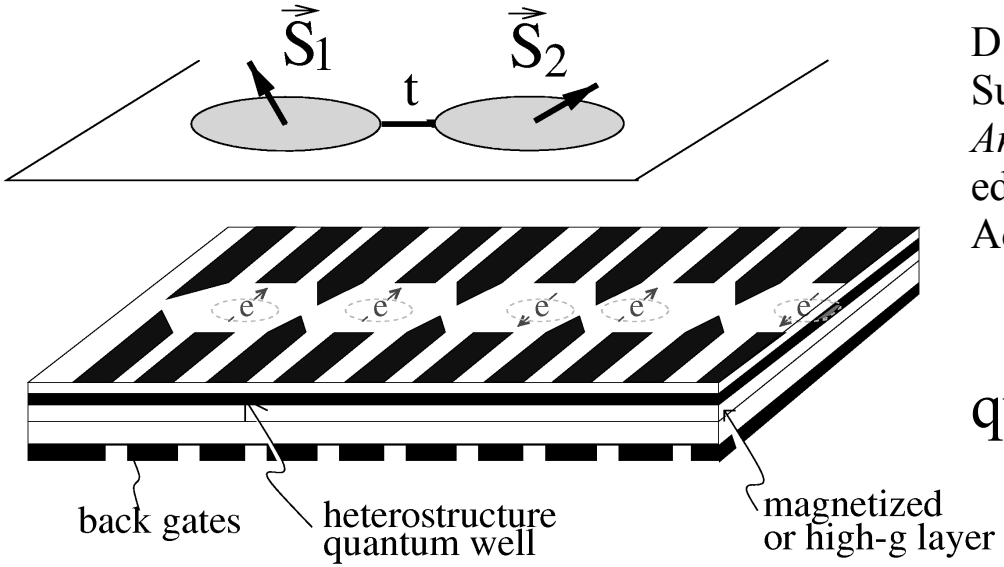
Computing on two conjoined encoded qubits

turn on extra exchange interactions between all 8 to make the supercoherent n=8 Hamiltonian

compute using additional exchanges

(respect gap condition)

Spintronic Quantum Computer



D.P. DiVincenzo, G. Burkard, D. Loss, E.V. Sukhorukov, in *Quantum Mesoscopic Phenomena And Mesoscopic Devices In Microelectronics,* eds. I.O. Kulik and R. Elliatioglu (NATO Advanced Study Institute, Turkey) 1999

qubit = spin of single electron

exchange interaction between qubits used for two-qubit interaction: tuned by raising and lowering barrier between dots to change overlap.

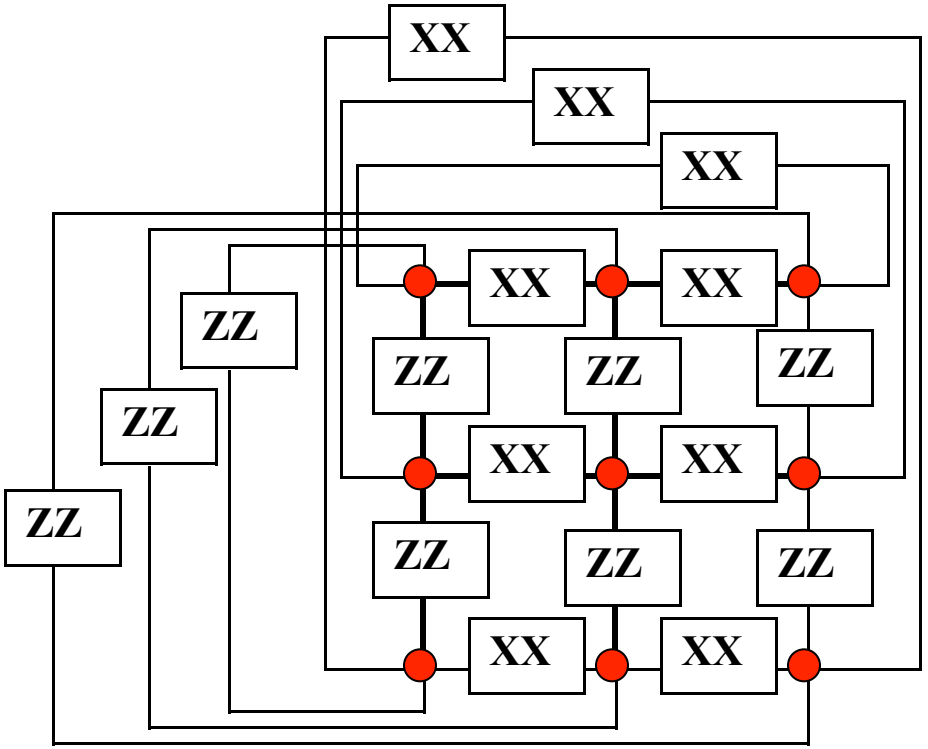
coupling strength: on ≈0.1 meV, off many orders of magnitude smaller

at temperatures below ≈1 K supercoherence

difficulty of geometry for n=8

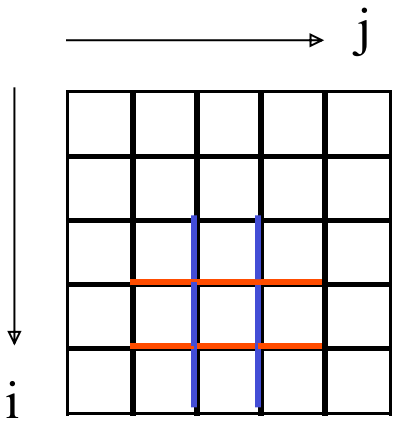
- combine vertical and lateral coupling schemes?

“As long as you’re going to think anyway, think big” - Donald Trump



Ground state is a quantum error correcting code (non-additive)

encodes 1 qubit of information in a single error detecting code



n by n Rectangular Toric lattice with qubits on vertices

XX coupling between qubits on adjacent columns, same row

ZZ coupling between qubits on adjacent rows, same column

$\boldsymbol{M}_{i,j} = \boldsymbol{M}$ acting on the *i*th row, *j*th column tensored with identity on all others

row operators $\boldsymbol{R}_i^{\boldsymbol{M}} = \prod_{j=1}^{n} \boldsymbol{M}_{i,j}$ column operators $\boldsymbol{C}_i^{\boldsymbol{M}} = \prod_{j=1}^{n} \boldsymbol{M}_{j,i}$

Stabilizer: double row $\boldsymbol{Z}$ operators $\boldsymbol{R}_a^{\boldsymbol{Z}} \boldsymbol{R}_b^{\boldsymbol{Z}}$, double column $\boldsymbol{X}$ operators $\boldsymbol{C}_a^{\boldsymbol{X}} \boldsymbol{C}_b^{\boldsymbol{X}}$ $(a \neq b)$

$2(n-1)$ stabilizer generators $\rightarrow$ $n^2 - 2(n-1) = (n-1)^2 + 1$ encoded qubits

$\bar{\boldsymbol{Z}}_{i,j} = \boldsymbol{Z}_{1,j} \boldsymbol{Z}_{i,j}$ $\bar{\boldsymbol{X}}_{i,j} = \boldsymbol{X}_{i,1} \boldsymbol{X}_{i,j}$ $(2 < i, j < n)$

$\bar{\boldsymbol{z}} = \boldsymbol{C}_1^{\boldsymbol{Z}}$ $\bar{\boldsymbol{x}} = \boldsymbol{R}_1^{\boldsymbol{X}}$

Hamiltonian=$\boldsymbol{H}$=sum of products of $\bar{\boldsymbol{Z}}_{i,j}, \bar{\boldsymbol{X}}_{i,j}$, stabilizer elements

encode information into $\bar{z}$ qubit

open question for $n > 5$: degeneracy of ground state, structure of spectrum

Instant Replay

- general idea: make decoherence mechanisms run uphill
- two qubit interactions used to construct a ground state for which any single qubit operation takes the system to a higher energy level (supercoherence)
- two qubit interactions understood using Pauli stabilizer symmetry
- exchange supercoherence allows for a fully universal quantum computer with realistic interactions and (pseudo-realistic) geometry
- quantum error correcting ground state

Work done in collaboration with

Professor K. B. Whaley (U. C. Berkeley, Chemistry)

Kenneth Brown (graduate student, U. C. Berkeley, Chemistry)

"Supercoherent Quantum Bits", Bacon, Brown, and Whaley quant-ph/0012018 (2000)

Thoughts I Think I Think

- finite bit simulations of entanglement (Brassard, Cleve, Tapp)

- continuous time versions of quantum algorithms (Farhi, Gutmann)

- noiseless (decoherence-free) subspaces, subsystems (Zanardi, Lidar, etc.)

- applications of "the ideas of information" to physics (opposite of quantum computation?)

Dave Bacon

visiting Institute for Quantum Information

Jan.17 - Feb. 14, April 9 - April 27

Office: 305 Steele

contact: dabacon@holmium.cchem.berkeley.edu

tidbit of the day: Is "laws of physics same in all reference frames" equivalent to "value of a bit the same in all reference frames"?